

Code: Program 1.sas

PROJECT: CLINICAL TRIAL DATA PIPELINE
(RAW -> SDTM -> ADaM -> TLF)
PROTOCOL: CLIN001
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/* 1. Create Mock Raw Demographics (RAW_DEMO) */
data raw_demo;
length subjid $10 site $5 sex $10 race $20 brthdt $10 rfstdt $10 trt $15;
infile datalines dlm=' ';
input subjid $ site $ sex $ race $ brthdt $ rfstdt $ trt $;
datalines;
101,S01,Male,Asian,15/05/1990,01/06/2024,Drug A
102,S01,Female,Asian,22/11/1985,05/06/2024,Placebo
103,S02,M,White,10/08/1978,12/06/2024,Drug A
104,S02,Female,Black,03/03/1995,15/06/2024,Placebo
105,S03,F,Asian,28/02/1982,20/06/2024,Drug A
;
run;

proc print data=raw_demo;
title "Raw Demographics Dataset";
run;

/* 2. Map Raw Data into SDTM DM */
data sdtm_dm;
set raw_demo;

/* Study and Unique Identifiers */
length studyid $10 usubjid $25 arm $20;
studyid = "CLIN001";
usubjid = catx('-', studyid, site, subjid);
arm = trt;

/* Convert raw date (DD/MM/YYYY) to SAS Date, then ISO 8601 string */
birth_num = input(brthdt, ddmmyy10.);
rfst_num = input(rfstdt, ddmmyy10.);

length rfstdc $10;
rfstdc = put(rfst_num, yymmdd10.);

/* Calculate Age using YRDIF */
age = floor(yrdif(birth_num, rfst_num, 'actual'));
length ageu $10;
ageu = "YEARS";

/* Standardize Sex to M / F */
length sex_std $1;
if upcase(substr(sex, 1, 1)) = 'M' then sex_std = 'M';
else if upcase(substr(sex, 1, 1)) = 'F' then sex_std = 'F';
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/* Standardize Race to uppercase */
race_std = upcase(race);

/* Drop raw variables and rename cleaned ones */
drop sex race birth_num rfst_num brthdt rfstdt site trt;
rename sex_std = sex;
race_std = race;
run;

proc print data=sdtm_dm;
title "SDTM DM (Demographics Domain)";
run;

/* 3. Build ADaM ADSL from SDTM DM */
data adam_adsl;
set sdtm_dm;

/* Planned Treatment Variable */
length trt01p $20;
trt01p = arm;

/* Analysis Categorization: Age Group */
length agegr1 $10;
if age < 40 then agegr1 = "<40";
else if age >= 40 then agegr1 = ">=40";

/* Population Flags */
length saffl $1;
if not missing(rfstdc) then saffl = "Y";
else saffl = "N";
run;

proc print data=adam_adsl;
title "ADaM ADSL (Subject-Level Analysis Dataset)";
run;

/* 4. Create Mock Raw Adverse Events (RAW_AE) */
data raw_ae;
length subjid $10 site $5 aeterm $40 aestdt $10 aeenddt $10 aesev $10 aerel $15;
infile datalines dlm=' ';
input subjid $ site $ aeterm $ aestdt $ aeenddt $ aesev $ aerel $;
datalines;
101,S01,mild headache,05/06/2024,07/06/2024,Mild,Possible
101,S01,Nausea,10/06/2024,,Moderate,Probable
102,S01,dizziness,08/06/2024,08/06/2024,MILD,Not Related
103,S02,FATIGUE,14/06/2024,18/06/2024,Severe,Related
104,S02,cough,20/06/2024,25/06/2024,Mild,Not Related
;
run;

proc print data=raw_ae;
title "Raw Adverse Events Dataset";
run;
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/* 5. Map Raw AE to SDTM AE */
/* Sort RAW_AE before using BY SUBJID */
proc sort data=raw_ae;
by subjid;
run;

data sdtm_ae;
set raw_ae;
by subjid;

/* Subject and Study Identifiers */
length studyid $10 usubjid $25;
studyid = "CLIN001";
usubjid = catx('-', studyid, site, subjid);

/* Sequence Number per Subject */
retain aeseq;
if first.subjid then aeseq = 1;
else aeseq + 1;

/* Standardize Dates to ISO 8601 */
st_num = input(aestdt, ddmmyy10.);
end_num = input(aeenddt, ddmmyy10.);
length aestdct aeendtc $10;
if not missing(st_num) then aestdct = put(st_num, yymmdd10.);
if not missing(end_num) then aeendtc = put(end_num, yymmdd10.);

/* Preferred Term */
length aedecod $40;
aedecod = upcase(strip(aeterm));

/* Standardize Severity */
length aesev_std $10;
aesev_std = upcase(strip(aesev));

/* Standardize Causality */
length aerele_std $15;
aerele_std = upcase(strip(aerele));

keep studyid usubjid aeseq aeterm aedecod aestdct aeendtc aesev_std aerele_std;
rename aesev_std = aesev
aerele_std = aerele;
run;

proc print data=sdtm_ae;
title "SDTM AE (Adverse Events Domain)";
run;

/* 6. Merge SDTM AE with ADSL to create ADaM ADAE */
proc sort data=sdtm_ae;
by usubjid;
run;

proc sort data=adam_adsl;
by usubjid;
run;
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data adam_adae;
merge sdm_ae(in=a)
adam_adsl(keep=usubjid trt01p rfstdc age agegr1 sex in=b);
by usubjid;
if a;

/* Actual Treatment */
length trta $20;
trta = trt01p;

/* Treatment Emergent Analysis Flag (TRTEMFL) */
length trtemfl $1;
if not missing(aestdtc) and not missing(rfstdc) then do;
if input(aestdtc, yymmdd10.) >= input(rfstdc, yymmdd10.) then trtemfl = "Y";
else trtemfl = "N";
end;
else trtemfl = "N";
run;

proc print data=adam_adae;
title "ADaM ADAE (Adverse Events Analysis Dataset)";
run;

/* STEP 7: GENERATE SAFETY SUMMARY TABLE (TLF) */
title2 "Safety Population";
proc freq data=adam_adae;
tables trta * aesev / norow nocol nopercnt;
run;

/* Patient Listing of All Severe Adverse Events */
title1 "Listing 16.2.7: Subjects with Severe Adverse Events";
proc print data=adam_adae noobs;
where aesev = 'SEVERE';
var usubjid trta aedecod aestdtc aesev aere1;
run;
title;
```